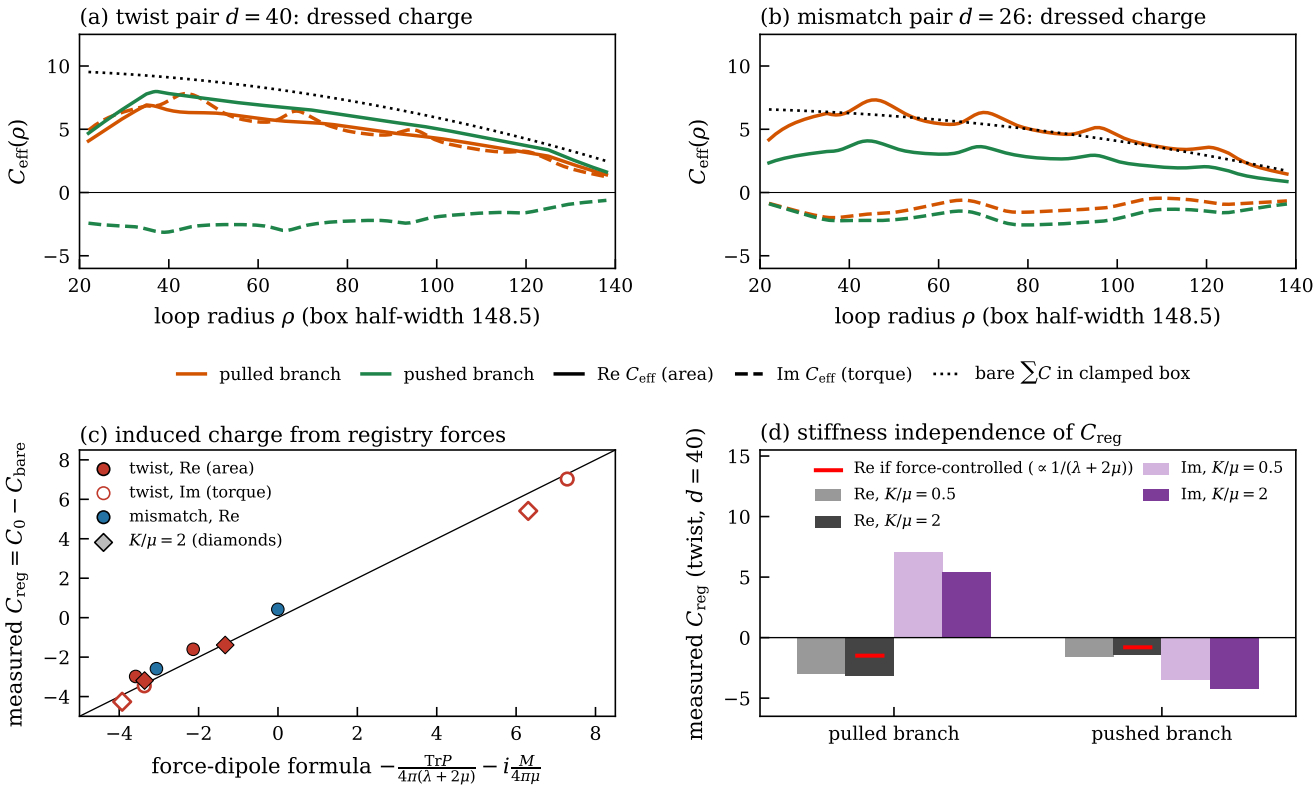
(a) twist pair $d = 40$: dressed charge
$C_{\mathrm{eff}}(\rho)$
loop radius ρ (box half-width 148.5)
(b) mismatch pair $d = 26$: dressed charge
$C_{\mathrm{eff}}(\rho)$
loop radius ρ (box half-width 148.5)
pulled branch
pushed branch
Re C_{eff} (area)
Im C_{eff} (torque)
bare $\sum C$ in clamped box
(c) induced charge from registry forces
twist, Re (area)
twist, Im (torque)
mismatch, Re
$K/\mu = 2$ (diamonds)
measured $C_{\mathrm{reg}} = C_0 - C_{\mathrm{bare}}$
force-dipole formula $-\frac{\mathrm{Tr}P}{4\pi(\lambda+2\mu)} - i\frac{M}{4\pi\mu}$
(d) stiffness independence of C_{reg}
Re if force-controlled ($\propto 1/(\lambda+2\mu)$)
Re, $K/\mu = 0.5$
Re, $K/\mu = 2$
Im, $K/\mu = 0.5$
Im, $K/\mu = 2$
measured C_{reg} (twist, $d = 40$)
pulled branch
pushed branch